

Section 6.2

Orthogonal Sets

Orthogonal Sets

A set of vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an orthogonal set if each pair of distinct vectors in the set is orthogonal; that is, if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

Theorem

If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

Orthogonal Bases

An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Theorem

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. Then, each $\mathbf{y}$ in W can be written as

$$\mathbf{y} = c_1\mathbf{u}_1 + \dots + c_p\mathbf{u}_p$$

where the weights are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

Example 1

Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$. Then express $\mathbf{x}$ as a linear combination of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} 8 \\ -4 \\ -3 \end{bmatrix}$$

Orthogonal Projections

Goal: Decompose a vector $\mathbf{y}$ in $\mathbb{R}^n$ into the sum of two vectors, one of which is a multiple of some nonzero vector $\mathbf{u}$ in $\mathbb{R}^n$ and the other orthogonal to $\mathbf{u}$.

That is, we wish to write

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} = \alpha \mathbf{u} + \mathbf{z}$$

where $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α and $\mathbf{z}$ is orthogonal to $\mathbf{u}$.

Orthogonal Projections

$$\mathbf{y} - \alpha \mathbf{u} = \mathbf{z}$$

We need $\mathbf{z}$ to be orthogonal to $\mathbf{u}$. Thus,

$$0 = (\mathbf{y} - \alpha \mathbf{u}) \cdot \mathbf{u}$$

$$0 = \mathbf{y} \cdot \mathbf{u} - \alpha \mathbf{u} \cdot \mathbf{u}$$

$$\alpha \mathbf{u} \cdot \mathbf{u} = \mathbf{y} \cdot \mathbf{u}$$

$$\alpha = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$$

Therefore,

$$\hat{\mathbf{y}} = \alpha \mathbf{u} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \cdot \mathbf{u}$$

Orthogonal Projections

If $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ where $\hat{\mathbf{y}} = \alpha \mathbf{u}$, then

- $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$
- $\mathbf{z}$ is the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$

If L is the subspace spanned by $\mathbf{u}$, $\hat{\mathbf{y}}$ is also called the orthogonal projection of $\mathbf{y}$ onto L .

Specifically,

$$\hat{\mathbf{y}} = \text{proj}_L \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

Example 2

Let $\mathbf{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$.

Write $\mathbf{y}$ as the sum of two orthogonal vectors, one in $\text{Span}\{\mathbf{u}\}$ and one orthogonal to $\mathbf{u}$.

Example 3

Let $\mathbf{y} = \begin{bmatrix} -3 \\ 9 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Compute the distance from $\mathbf{y}$ to the line through $\mathbf{u}$ and the origin.

Orthonormal Sets

A set $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal set if it is an orthogonal set of unit vectors.

If $W = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$, then $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for W .

Matrices with Orthonormal Columns

An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

Matrices with Orthonormal Columns

Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}$ and $\mathbf{y}$ be vectors in $\mathbb{R}^n$. Then,

- a) $\|U\mathbf{x}\| = \|\mathbf{x}\|$
- b) $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$
- c) $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$

Orthogonal and Unitary Matrices

An orthogonal matrix is a square matrix with real entries and orthonormal columns; that is, a square matrix where $U^T = U^{-1}$.

A unitary matrix is a square matrix with complex entries and orthonormal columns.

Example 4

Determine if each statement is true or false.
Assume all vectors are in $\mathbb{R}^n$.

- a) Every linearly independent set in $\mathbb{R}^n$ is an orthogonal set.
- b) If $\mathbf{y}$ is a linear combination of nonzero vectors from an orthogonal set, then the weights in the linear combination can be computed without row operations on a matrix.

Example 4

Determine if each statement is true or false.
Assume all vectors are in $\mathbb{R}^n$.

- c) If the vectors in an orthogonal set of nonzero vectors are normalized, then some of the new vectors may not be orthogonal.
- d) A matrix with orthonormal columns is an orthogonal matrix.

Example 5

Suppose W is a subspace of $\mathbb{R}^n$ spanned by n nonzero orthogonal vectors. Explain why $W = \mathbb{R}^n$.
